

Overview of Windows Operating System

1. What is the primary function of an operating system?

- A) Running applications
- B) Managing hardware resources
- C) Internet browsing
- D) Data analysis
- Answer: B) Managing hardware resources

2. Which of the following is a feature of Windows Operating System?

- A) Task Manager
- B) Siri
- C) Time Machine
- D) Terminal
- Answer: A) Task Manager

3. Which file system is commonly used in Windows 10?

- A) ext4
- B) NTFS
- C) HFS+
- D) FAT32
- Answer: B) NTFS

4. What is the purpose of the Windows Registry?

- A) To store user settings
- B) To manage RAM
- C) To install new hardware
- D) To store system configuration information
- Answer: D) To store system configuration information

5. What is the default web browser in Windows 10?

- A) Internet Explorer
- B) Safari
- C) Google Chrome
- D) Microsoft Edge
- Answer: D) Microsoft Edge

6. Which utility is used for disk defragmentation in Windows?

- A) Disk Cleanup
- B) Disk Defragmenter
- C) CHKDSK

- D) Task Manager
- Answer: B) Disk Defragmenter

7. **What is Cortana in Windows 10?**

- A) A file manager
- B) A digital assistant
- C) An antivirus program
- D) A web browser
- Answer: B) A digital assistant

8. **Windows Update is used for what purpose?**

- A) Cleaning the disk
- B) Installing new software
- C) Updating Windows software
- D) Backing up files
- Answer: C) Updating Windows software

9. **Which feature allows multiple desktops in Windows 10?**

- A) Task View
- B) Start Menu
- C) Control Panel
- D) Action Center
- Answer: A) Task View

10. **What is the command line interface in Windows called?**

- A) Terminal
- B) Console
- C) Command Prompt
- D) Shell
- Answer: C) Command Prompt

Hyper-V

1. **What is Hyper-V primarily used for?**

- A) Disk management
- B) Creating virtual machines
- C) Internet browsing
- D) Text editing
- Answer: B) Creating virtual machines

2. **Which edition of Windows includes Hyper-V?**

- A) Home
- B) Professional
- C) Starter
- D) Basic
- Answer: B) Professional

3. **Hyper-V virtual machines are isolated from each other using what technology?**

- A) Firewalls
- B) Virtual Switches
- C) Network Bridges
- D) NAT
- Answer: B) Virtual Switches

4. **What is a 'Snapshot' in Hyper-V?**

- A) A backup of a VM
- B) A screenshot of a VM
- C) A detailed report of VM performance
- D) A quick restore point
- Answer: D) A quick restore point

5. **Hyper-V supports which type(s) of virtualization?**

- A) Hardware
- B) Software
- C) Both hardware and software
- D) Neither hardware nor software
- Answer: C) Both hardware and software

6. **In Hyper-V, what is a 'Virtual Switch'?**

- A) A physical switch for managing VMs
- B) A software-based networking device
- C) A tool for switching between VMs
- D) A hardware component in the host machine
- Answer: B) A software-based networking device

7. **Which feature in Hyper-V allows VMs to access external networks?**

- A) NAT
- B) Virtual LAN
- C) External Virtual Switch
- D) Bridged networking
- Answer: C) External Virtual Switch

8. **Can Hyper-V run on Windows 10 Home Edition?**

- A) Yes, by default
- B) No, it's not available
- C) Yes, with an upgrade
- D) Only with third-party software
- Answer: B) No, it's not available

9. **What is the minimum requirement of RAM for installing Hyper-V on a Windows machine?**

- A) 1 GB
- B) 2 GB
- C) 4 GB
- D) 8 GB
- Answer: C) 4 GB

10. **Hyper-V can be used to run which type of operating systems?**

- A) Only Windows
- B) Only Linux
- C) Both Windows and Linux
- D) Neither Windows nor Linux
- Answer: C) Both Windows and Linux

Disk in Windows

1. **What is a primary function of the Disk Management tool in Windows?**

- A) Installing new applications
- B) Managing disk partitions
- C) Updating Windows OS
- D) Managing network connections
- Answer: B) Managing disk partitions

2. **Which file system is NOT supported by Windows Disk Management?**

- A) NTFS
- B) FAT32
- C) exFAT
- D) ext4
- Answer: D) ext4

3. **What is the maximum size of a hard disk that MBR (Master Boot Record) can support?**

- A) 2 TB
- B) 4 TB

- C) 16 TB
- D) 1 TB
- Answer: A) 2 TB

4. **What does the term 'Disk Defragmentation' refer to in Windows?**

- A) Deleting unnecessary files
- B) Organizing files to contiguous sectors
- C) Encrypting disk data
- D) Expanding disk capacity
- Answer: B) Organizing files to contiguous sectors

5. **What is a characteristic of a Dynamic Disk in Windows?**

- A) Can only have primary partitions
- B) Can create volumes that span multiple disks
- C) Does not support RAID configurations
- D) Limited to 2 TB disk size
- Answer: B) Can create volumes that span multiple disks

6. **Which of the following is NOT a type of disk partition in Windows?**

- A) Primary
- B) Extended
- C) Logical
- D) Virtual
- Answer: D) Virtual

7. **What is a GPT (GUID Partition Table) in the context of Windows disk management?**

- A) A type of file system
- B) A disk partition style
- C) A disk defragmentation tool
- D) A type of disk encryption
- Answer: B) A disk partition style

8. **Which tool in Windows can be used to check a disk for file system errors?**

- A) Disk Cleanup
- B) Disk Management
- C) CHKDSK
- D) Device Manager
- Answer: C) CHKDSK

9. **What does converting a basic disk to a dynamic disk in Windows do?**

- A) Increases the disk capacity

- B) Improves the disk's read/write speed
- C) Allows creation of advanced volumes like spanned and striped volumes
- D) Changes the file system to NTFS
- Answer: C) Allows creation of advanced volumes like spanned and striped volumes

10. Which of the following is NOT a valid reason for a disk to show as 'Unallocated' in Windows Disk Management?

- A) The disk is damaged
- B) The disk has a non-Windows file system
- C) The disk is fully partitioned and formatted
- D) The disk is new and not initialized
- Answer: C) The disk is fully partitioned and formatted

1. What is the maximum size for a volume using the FAT32 file system in Windows?

- A. 2 TB
- B. 32 GB
- C. 4 GB
- D. 16 TB
- Answer: B. 32 GB

2. Which tool in Windows is primarily used for partitioning hard drives?

- A. Device Manager
- B. Disk Cleanup
- C. Disk Management
- D. Disk Defragmenter
- Answer: C. Disk Management

3. What is the purpose of the 'Shrink Volume' option in Disk Management?

- A. To delete a volume
- B. To reduce the size of a volume
- C. To format a volume
- D. To expand the size of a volume
- Answer: B. To reduce the size of a volume

4. Which file system introduced with Windows NT supports large volumes and advanced features like file permissions?

- A. FAT32
- B. NTFS
- C. exFAT

- D. FAT16
- Answer: B. NTFS

5. What is the main advantage of the exFAT file system over FAT32?

- A. Better security
- B. Support for larger files and volumes
- C. Faster speed
- D. Compatibility with older systems
- Answer: B. Support for larger files and volumes

6. Which Windows utility is used to check the disk for file system errors and bad sectors?

- A. Disk Cleanup
- B. CHKDSK
- C. Disk Defragmenter
- D. Disk Management
- Answer: B. CHKDSK

7. In Windows, what does it mean to 'mount' a volume?

- A. To format a hard drive
- B. To assign a drive letter or path to a volume
- C. To partition a hard drive
- D. To compress a hard drive
- Answer: B. To assign a drive letter or path to a volume

8. What is the primary function of the Disk Cleanup tool in Windows?

- A. Defragmenting the disk
- B. Repairing disk errors
- C. Freeing up disk space
- D. Partitioning the disk
- Answer: C. Freeing up disk space

9. In Windows, what is the default file system used for formatting USB flash drives?

- A. NTFS
- B. FAT32
- C. exFAT
- D. FAT16
- Answer: B. FAT32

10. What feature of NTFS allows users to encrypt files and folders on a Windows volume?

- A. BitLocker
- B. EFS (Encrypting File System)

- C. Windows Defender
- D. Security Essentials
- Answer: B. EFS (Encrypting File System)

Active Directory (AD)

1. What is Active Directory?

- A) A database system
- B) A network protocol
- C) A directory service
- D) A file sharing service
- Answer: C) A directory service

2. Which protocol does Active Directory use for directory services?

- A) HTTP
- B) SMTP
- C) LDAP
- D) FTP
- Answer: C) LDAP

3. What is a Global Catalog in Active Directory?

- A) A user account
- B) A full listing of all AD objects
- C) A partial listing of all AD objects
- D) A security group
- Answer: C) A partial listing of all AD objects

4. What are Group Policies in Active Directory?

- A) Settings for user desktops
- B) Password policies
- C) Centralized configuration
- D) All of the above
- Answer: D) All of the above

5. What is the main purpose of Organizational Units (OUs) in Active Directory?

- A) Group computers based on location
- B) Group objects for delegation
- C) Store user accounts
- D) All of the above

- **Answer: B) Group objects for delegation**

6. **Which of the following is a function of Active Directory?**

- A) Email delivery
- B) Web hosting
- C) Identity management
- D) Data analysis
- **Answer: C) Identity management**

7. **What is a Trust in Active Directory?**

- A) A security group
- B) A backup protocol
- C) A relationship between domains
- D) A type of user account
- **Answer: C) A relationship between domains**

8. **Which of the following is a feature of Active Directory?**

- A) Data compression
- B) Multimedia streaming
- C) Software deployment
- D) Graphics rendering
- **Answer: C) Software deployment**

9. **How is DNS integrated with Active Directory?**

- A) For email routing
- B) For resolving domain names to IP addresses
- C) For file sharing
- D) It is not integrated
- **Answer: B) For resolving domain names to IP addresses**

10. **What is an Active Directory Schema?**

- A) A group policy
- B) A set of rules for object creation
- C) A network topology
- D) A security protocol
- **Answer: B) A set of rules for object creation**

Active Directory Domain Services (ADDS)

1. **What is the primary role of Active Directory Domain Services (ADDS)?**

- A) Web hosting

- B) Database management
- C) User and resource management in a domain
- D) Email services
- **Answer: C) User and resource management in a domain**

2. Which feature of ADDS allows for central management of user data and security?

- A) File sharing
- B) Directory services
- C) Web services
- D) DHCP services
- **Answer: B) Directory services**

3. What is a forest in ADDS?

- A) A collection of trees
- B) A single domain controller
- C) A group of users
- D) A backup system
- **Answer: A) A collection of trees**

4. How does ADDS provide security for domain data?

- A) Through encryption
- B) Through firewalls
- C) Through access control lists
- D) Through antivirus software
- **Answer: C) Through access control lists**

5. What is the main purpose of a domain in ADDS?

- A) To provide a web service
- B) To segment network traffic
- C) To organize and manage network resources
- D) To provide email services
- **Answer: C) To organize and manage network resources**

6. Which feature of ADDS allows administrators to apply policies to multiple users or computers at once?

- A) Group Policy
- B) OU
- C) Global Catalog
- D) Schema
- **Answer: A) Group Policy**

7. What is the primary use of a Global Catalog in ADDS?

- A) To store detailed information about every object in the domain
- B) To store a partial set of attributes for every object in the forest
- C) To act as a primary domain controller
- D) To provide email services
- **Answer: B) To store a partial set of attributes for every object in the forest**

8. Which protocol is primarily used by ADDS for directory access and management?

- A) HTTP
- B) FTP
- C) SMTP
- D) LDAP
- **Answer: D) LDAP**

9. What is a site in the context of ADDS?

- A) A web page
- B) A physical or logical group of domain controllers
- C) A user group
- D) A security protocol
- **Answer: B) A physical or logical group of domain controllers**

10. What is the function of a Read-Only Domain Controller (RODC) in ADDS?

- A) To provide full write access to the directory
- B) To provide read-only access to the directory in specific scenarios
- C) To serve as a backup domain controller
- D) To host web services
- **Answer: B) To provide read-only access to the directory in specific scenarios**

Domain Controller (DC)

1. What is the primary role of a Domain Controller in a network?

- A) To provide file sharing services
- B) To authenticate and authorize users and computers in a domain
- C) To distribute IP addresses
- D) To serve web pages
- **Answer: B) To authenticate and authorize users and computers in a domain**

2. In an Active Directory environment, what is the main function of a Domain Controller?

- A) To manage web services
- B) To host databases
- C) To store and manage directory information

- D) To provide email services
- **Answer: C) To store and manage directory information**

3. **Which service must be installed on a server to promote it to a Domain Controller?**

- A) IIS
- B) DHCP
- C) DNS
- D) Active Directory Domain Services (ADDS)
- **Answer: D) Active Directory Domain Services (ADDS)**

4. **How does a Domain Controller facilitate logins in a Windows domain?**

- A) By providing a user interface
- B) By managing web sessions
- C) By authenticating user credentials
- D) By distributing software
- **Answer: C) By authenticating user credentials**

5. **What happens if a Domain Controller fails in a multi-DC environment?**

- A) The entire network crashes
- B) Another Domain Controller can take over its duties
- C) Users can no longer log in
- D) The internet connection is lost
- **Answer: B) Another Domain Controller can take over its duties**

6. **Which feature of a Domain Controller allows for policy management across a network?**

- A) Group Policy
- B) File Sharing
- C) DHCP
- D) DNS
- **Answer: A) Group Policy**

7. **What is the primary database stored on a Domain Controller?**

- A) SQL Database
- B) Active Directory Database
- C) Web Database
- D) DHCP Database
- **Answer: B) Active Directory Database**

8. **What is the purpose of having multiple Domain Controllers in a domain?**

- A) To increase web hosting capacity
- B) To provide load balancing and redundancy

- C) To store more files
- D) To increase email capacity
- **Answer: B) To provide load balancing and redundancy**

9. **In which scenario would you use a Read-Only Domain Controller (RODC)?**

- A) In high-security areas where read-write access is not required
- B) In scenarios where high-speed internet is available
- C) In small offices with limited users
- D) In scenarios where ADDS is not required
- **Answer: A) In high-security areas where read-write access is not required**

10. **What role does a Domain Controller play in Active Directory replication?**

- A) It does not participate in replication
- B) It acts as a replication source and destination
- C) It only receives replication data
- D) It only sends replication data
- **Answer: B) It acts as a replication source and destination**

Additional Domain Controller

1. **What is the primary purpose of an Additional Domain Controller in a network?**

- A. To provide fault tolerance
- B. To serve as the only domain controller
- C. To support non-Windows clients
- D. To act as a standalone server
- **Answer: A**

2. **Which feature is automatically enabled in an Additional Domain Controller in a Windows domain?**

- A. DHCP server
- B. Active Directory replication
- C. File Sharing
- D. Internet Information Services (IIS)
- **Answer: B**

3. **In a domain with multiple domain controllers, what happens if the Primary Domain Controller (PDC) fails?**

- A. The domain stops functioning
- B. An Additional Domain Controller can be promoted to PDC
- C. User authentication fails
- D. All computers lose network connectivity

- **Answer: B**

4. **What is the process of synchronizing directory information across domain controllers called?**

- A. Directory Synchronization
- B. Replication
- C. Data Mirroring
- D. Data Sharing
- **Answer: B**

5. **What does an Additional Domain Controller use to authenticate and update information in a domain?**

- A. Windows Registry
- B. Active Directory
- C. Local Database
- D. Network File System
- **Answer: B**

6. **When deploying an Additional Domain Controller, which protocol is primarily used for replication?**

- A. HTTP
- B. FTP
- C. SMTP
- D. Kerberos
- **Answer: D**

7. **Can an Additional Domain Controller also provide DNS services in a domain?**

- A. Yes
- B. No
- C. Only if it is the PDC
- D. Only in absence of a primary DNS server
- **Answer: A**

8. **Which feature ensures that an Additional Domain Controller has an updated copy of the Active Directory database?**

- A. Live Migration
- B. Replication
- C. Backup and Restore
- D. Load Balancing
- **Answer: B**

9. **In a Windows domain, what role does an Additional Domain Controller primarily play?**

- A. Backup server
- B. Web server

- C. File server
- D. Authentication server
- **Answer: D**

10. **What is a recommended practice when deploying an Additional Domain Controller?**

- A. Place it in the same physical location as the PDC
- B. Use a different operating system than the PDC
- C. Ensure it is part of a different domain
- D. Ensure it is in a different site for disaster recovery purposes
- **Answer: D**

Member Server

1. **What is a Member Server in a Windows domain?**

- A. A server that hosts the Active Directory
- B. A server that is part of a domain but does not host Active Directory
- C. A standalone server not part of any domain
- D. A server that acts as a client
- **Answer: B**

2. **Which of the following is a role that can be assigned to a Member Server?**

- A. Domain Controller
- B. DNS Server
- C. DHCP Server
- D. Both B and C
- **Answer: D**

3. **Can a Member Server provide network services like file sharing and printing?**

- A. Yes
- B. No
- C. Only if it is also a Domain Controller
- D. Only if authorized by a Domain Controller
- **Answer: A**

4. **What is a key difference between a Domain Controller and a Member Server?**

- A. Member Server cannot have local users and groups
- B. Member Server does not participate in Active Directory replication
- C. Member Server cannot connect to the internet
- D. Member Server cannot run server applications
- **Answer: B**

5. Which security practice is commonly applied to a Member Server in a domain?

- A. Using a different operating system than other servers in the domain
- B. Applying domain-level security policies
- C. Disabling all network interfaces
- D. Operating in a standalone mode
- **Answer: B**

6. What must be done to join a server as a Member Server in a domain?

- A. Format the server
- B. Change the server's IP address
- C. Install a specific Member Server software
- D. Add the server to the domain through system properties
- **Answer: D**

7. Can a Member Server in a Windows domain also function as a web server using IIS?

- A. Yes
- B. No
- C. Only with special permissions from the Domain Controller
- D. Only if it is not providing any other services
- **Answer: A**

8. What is required for a server to be managed as a Member Server in a domain?

- A. It must have a static IP address
- B. It must run a server version of Windows
- C. It must be physically located in the same building as the Domain Controller
- D. It must be joined to the domain
- **Answer: D**

9. Which of the following can be a benefit of using a Member Server in a domain?

- A. Centralized management
- B. Automatic backup of all data
- C. No need for network security
- D. Independence from domain policies
- **Answer: A**

10. When would a Member Server typically be used over a Domain Controller?

- A. When a server is needed to store user data
- B. When a new domain is being created
- C. When high-level domain management is required
- D. When the network does not have any other servers

- **Answer:** A

DNS (Domain Name System)

1. What is the primary purpose of DNS in a network?

- A) Assigning IP addresses
- B) Translating domain names to IP addresses
- C) Blocking malicious websites
- D) Routing email traffic
- **Answer:** B) Translating domain names to IP addresses

2. Which type of DNS record maps a domain name to an IP address?

- A) MX record
- B) CNAME record
- C) A record
- D) PTR record
- **Answer:** C) A record

3. What is a DNS zone?

- A) A device that stores DNS records
- B) A protocol for secure DNS queries
- C) A distinct part of the domain namespace
- D) The process of updating DNS records
- **Answer:** C) A distinct part of the domain namespace

4. Which record is used for mail server identification?

- A) A record
- B) MX record
- C) CNAME record
- D) SRV record
- **Answer:** B) MX record

5. What is the purpose of a CNAME record?

- A) Connect a domain to an IP address
- B) Specify authoritative name servers
- C) Alias one name to another
- D) Reverse mapping of IP to domain
- **Answer:** C) Alias one name to another

6. What is DNS forwarding?

- A) Redirecting all DNS traffic to another server
- B) Updating DNS records automatically
- C) Forwarding emails via DNS
- D) Resolving external domain names
- **Answer:** A) Redirecting all DNS traffic to another server

7. **What does TTL in DNS stand for?**

- A) Total Transfer Length
- B) Time To Live
- C) Transfer Type Limit
- D) Transaction Time Log
- **Answer:** B) Time To Live

8. **What is a DNS resolver?**

- A) A server that hosts DNS records
- B) A client that requests DNS information
- C) A protocol for DNS queries
- D) A type of DNS record
- **Answer:** B) A client that requests DNS information

9. **What does an NS record in DNS do?**

- A) Points to a network service
- B) Identifies the name servers for a domain
- C) Specifies network subnet
- D) Indicates a network switch
- **Answer:** B) Identifies the name servers for a domain

10. **What is the function of a PTR record?**

- A) Maps a domain name to an IP address
- B) Points to a related DNS zone
- C) Reverse mapping of IP to domain
- D) Indicates priority of mail servers
- **Answer:** C) Reverse mapping of IP to domain

IPAM (IP Address Management)

1. **What is the primary function of IPAM?**

- A) Managing network devices
- B) Assigning domain names
- C) Monitoring network traffic

- D) Managing IP address space
- **Answer:** D) Managing IP address space

2. **Which feature is commonly included in IPAM solutions?**

- A) Email server configuration
- B) IP address tracking and inventory
- C) Physical server maintenance
- D) Website hosting
- **Answer:** B) IP address tracking and inventory

3. **What does DHCP integration in IPAM help with?**

- A) Email management
- B) Intrusion detection
- C) Automating IP address assignment
- D) Physical network cabling
- **Answer:** C) Automating IP address assignment

4. **How does IPAM aid in network planning?**

- A) By providing chat support
- B) Offering historical data and usage trends
- C) Selling network equipment
- D) Writing network policies
- **Answer:** B) Offering historical data and usage trends

5. **What is a key benefit of using IPAM in a large network?**

- A) Reducing email spam
- B) Enhancing physical security
- C) Improving IP address utilization
- D) Managing user accounts
- **Answer:** C) Improving IP address utilization

6. **Which of the following is a challenge that IPAM addresses?**

- A) Printer setup
- B) IP address conflicts
- C) Software installation
- D) Keyboard configuration
- **Answer:** B) IP address conflicts

7. **What role does IPAM play in network compliance?**

- A) Ensures ergonomic standards
- B) Manages software licenses

- C) Helps in adhering to IP allocation policies
- D) Oversees employee attendance
- **Answer:** C) Helps in adhering to IP allocation policies

8. **How does IPAM interact with DNS?**

- A) Manages domain sale
- B) Integrates IP address records with DNS
- C) Provides web hosting services
- D) Filters internet content
- **Answer:** B) Integrates IP address records with DNS

9. **What is a subnet in the context of IPAM?**

- A) A type of network cable
- B) A division of an IP address space
- C) A network routing protocol
- D) A server hardware model
- **Answer:** B) A division of an IP address space

10. **Why is IPAM important in cloud environments?**

- A) For managing virtual machine sizes
- B) For organizing email servers
- C) For dynamic IP address allocation
- D) For website design
- **Answer:** C) For dynamic IP address allocation

DHCP (Dynamic Host Configuration Protocol)

1. **What is the main function of DHCP?**

- A) Filtering web content
- B) Assigning IP addresses dynamically
- C) Providing DNS services
- D) Monitoring network performance
- **Answer:** B) Assigning IP addresses dynamically

2. **What is a DHCP lease?**

- A) A network cable type
- B) The duration for which an IP address is assigned
- C) A contract for network services
- D) A type of network switch
- **Answer:** B) The duration for which an IP address is assigned

3. Which component is essential in a DHCP setup?

- A) DHCP relay agent
- B) Network printer
- C) Web server
- D) File server
- **Answer:** A) DHCP relay agent

4. What happens when a DHCP lease expires?

- A) The network is disconnected
- B) The IP address is renewed or released
- C) The server shuts down
- D) Email services are interrupted
- **Answer:** B) The IP address is renewed or released

5. What is a DHCP reservation?

- A) Booking a time slot for network access
- B) Assigning a specific IP address to a specific device
- C) Reserving network bandwidth
- D) Scheduling server maintenance
- **Answer:** B) Assigning a specific IP address to a specific device

6. In DHCP, what is a scope?

- A) The physical range of the network
- B) The range of IP addresses available for assignment
- C) The type of network devices used
- D) The budget for network equipment
- **Answer:** B) The range of IP addresses available for assignment

7. What does a DHCP NAK message signify?

- A) Acknowledgment of a valid request
- B) Notification of an available IP address
- C) Rejection of a DHCP request
- D) Confirmation of a lease renewal
- **Answer:** C) Rejection of a DHCP request

8. How does DHCP help in network administration?

- A) By encrypting network traffic
- B) Automating IP address allocation
- C) Managing user passwords
- D) Monitoring internet usage

- **Answer:** B) Automating IP address allocation

9. **What is the first message sent by a client in the DHCP process?**

- A) DHCPREQUEST
- B) DHCP OFFER
- C) DHCPDISCOVER
- D) DHCPACK
- **Answer:** C) DHCPDISCOVER

10. **Which DHCP setting defines the network configuration for a client?**

- A) Lease time
- B) DHCP options
- C) Scope name
- D) Relay agent
- **Answer:** B) DHCP options

1. **What is the primary purpose of DHCP in a network?**

- A) To provide network security
- B) To manage network bandwidth
- C) To assign IP addresses to devices
- D) To route data packets
- **Answer:** C) To assign IP addresses to devices

2. **Which of the following is a benefit of using DHCP in a network?**

- A) Reduces the need for manual IP address configuration
- B) Increases network speed
- C) Provides end-to-end encryption
- D) Reduces network traffic
- **Answer:** A) Reduces the need for manual IP address configuration

3. **In DHCP, what is a scope?**

- A) A set of rules for assigning IP addresses
- B) A range of IP addresses available for assignment
- C) A specific IP address assigned to a client
- D) A protocol for securing IP address assignments
- **Answer:** B) A range of IP addresses available for assignment

4. **What is a DHCP lease?**

- A) A permanent IP address assignment
- B) A temporary IP address assignment

- C) A fixed bandwidth allocation
- D) A network security protocol
- Answer: B) A temporary IP address assignment

5. **What happens when a DHCP lease expires?**

- A) The IP address is permanently assigned to the device
- B) The device must request a new IP address
- C) The device is disconnected from the network
- D) The lease is automatically renewed
- Answer: B) The device must request a new IP address

6. **What is a DHCP reservation?**

- A) A permanent IP address assignment for a specific device
- B) A temporary allocation of network resources
- C) A high-priority lease for critical devices
- D) A security measure to prevent unauthorized access
- Answer: A) A permanent IP address assignment for a specific device

7. **What is the function of a DHCP relay agent?**

- A) To encrypt DHCP traffic
- B) To forward DHCP requests between subnets
- C) To assign static IP addresses
- D) To monitor DHCP traffic for security threats
- Answer: B) To forward DHCP requests between subnets

8. **How does DHCP prevent IP address conflicts?**

- A) By using encryption
- B) By assigning permanent IP addresses
- C) By keeping a database of assigned addresses
- D) By limiting the number of devices on the network
- Answer: C) By keeping a database of assigned addresses

9. **What is the first message sent by a client in the DHCP process?**

- A) DHCPREQUEST
- B) DHCPACK
- C) DHCPDISCOVER
- D) DHCPOFFER
- Answer: C) DHCPDISCOVER

10. **What is required to set up a DHCP server in a Windows environment?**

- A) A dedicated physical server

- B) Installation of DHCP server role
- C) A specific type of network cable
- D) A third-party software license
- Answer: B) Installation of DHCP server role

1. **What is the primary purpose of a VPN in a Windows network environment?**

- A) To increase internet speed
- B) To provide a secure connection over a public network
- C) To monitor network traffic
- D) To provide additional storage space
- Answer: B) To provide a secure connection over a public network

2. **What is DirectAccess in Windows?**

- A) A tool for remote desktop access
- B) A VPN alternative that provides seamless, automatic, and always-on remote access
- C) A method for compressing data on the network
- D) A protocol for encrypting emails
- Answer: B) A VPN alternative that provides seamless, automatic, and always-on remote access

3. **Which protocol is commonly used for securely transmitting VPN traffic?**

- A) HTTP
- B) SMTP
- C) IPSec
- D) FTP
- Answer: C) IPSec

4. **What is Network Address Translation (NAT) used for in a network?**

- A) To encrypt data packets
- B) To assign IP addresses to devices
- C) To allow multiple devices to share a single public IP address
- D) To monitor network traffic
- Answer: C) To allow multiple devices to share a single public IP address

5. **What is the function of a RADIUS server in a network?**

- A) To route data packets
- B) To provide remote access services
- C) To centralize authentication and accounting for network access
- D) To assign IP addresses
- Answer: C) To centralize authentication and accounting for network access

6. **What is required to implement a VPN in Windows Server?**

- A) Additional hardware routers
- B) A third-party VPN service
- C) Installation of the Remote Access server role
- D) A dedicated internet connection
- Answer: C) Installation of the Remote Access server role

7. **How does a firewall contribute to remote access solutions?**

- A) By increasing connection speed
- B) By acting as a DHCP server
- C) By providing data storage solutions
- D) By controlling incoming and outgoing network traffic
- Answer: D) By controlling incoming and outgoing network traffic

8. **What is the purpose of implementing a site-to-site VPN?**

- A) To connect individual users to a network
- B) To connect two or more remote networks securely over the internet
- C) To increase bandwidth
- D) To serve as a backup for data storage
- Answer: B) To connect two or more remote networks securely over the internet

9. **Which feature in Windows Server allows for the management of network policies and access?**

- A) Network Policy Server (NPS)
- B) Active Directory
- C) DHCP Server
- D) File Server Resource Manager
- Answer: A) Network Policy Server (NPS)

10. **What is a common use case for implementing a Remote Desktop Gateway (RD Gateway) in Windows?**

- A) To manage emails remotely
- B) To enhance website hosting
- C) To enable secure connections to internal network resources from outside the corporate network
- D) To provide additional storage
- Answer: C) To enable secure connections to internal network resources from outside the corporate network

Network Policy Server (NPS)

1. **What is the primary function of Network Policy Server (NPS) in Windows?**

- A) File sharing

- B) Network policy management and enforcement
- C) User authentication
- D) Data encryption
- Answer: B) Network policy management and enforcement

2. NPS in Windows acts as a _____.

- A) DHCP server
- B) RADIUS server
- C) DNS server
- D) Web server
- Answer: B) RADIUS server

3. Which of the following is a feature of NPS?

- A) Virtual machine management
- B) Network Access Protection (NAP)
- C) Active Directory management
- D) Firewall configuration
- Answer: B) Network Access Protection (NAP)

4. In NPS, what is a Connection Request Policy used for?

- A) Defining file sharing permissions
- B) Determining how incoming connection requests are processed
- C) Managing user group memberships
- D) Configuring VPN protocols
- Answer: B) Determining how incoming connection requests are processed

5. Which protocol is used by NPS for centralized authentication?

- A) SSH
- B) RADIUS
- C) LDAP
- D) Kerberos
- Answer: B) RADIUS

6. NPS can be used to authenticate and authorize users for which type of network access?

- A) Local network access only
- B) Internet access only
- C) VPN and wireless access
- D) Remote desktop access
- Answer: C) VPN and wireless access

7. Which of the following is NOT a role of NPS?

- A) RADIUS proxy

- B) RADIUS server
- C) DHCP server
- D) RADIUS client
- Answer: C) DHCP server

8. NPS integrates with which Windows service for user identity verification?

- A) Windows Defender
- B) Active Directory
- C) Windows Update
- D) Windows Firewall
- Answer: B) Active Directory

9. What is required to configure NPS as a RADIUS server?

- A) RADIUS clients
- B) Connection Request Policies
- C) Network Policies
- D) All of the above
- Answer: D) All of the above

10. In NPS, what does a Network Policy define?

- A) Network bandwidth limits
- B) Conditions and settings for network access
- C) IP address assignments
- D) Firewall rules
- Answer: B) Conditions and settings for network access

Local Policies

1. What is the purpose of Local Policies in Windows?

- A) Managing user accounts
- B) Configuring local computer settings
- C) Sharing files over a network
- D) Installing software applications
- Answer: B) Configuring local computer settings

2. Which tool is used to configure Local Policies in Windows?

- A) Control Panel
- B) Command Prompt
- C) Local Security Policy
- D) Registry Editor

- Answer: C) Local Security Policy

3. Under Local Policies, what does the Audit Policy determine?

- A) Password complexity requirements
- B) Types of events to be logged
- C) File sharing permissions
- D) Network access rights
- Answer: B) Types of events to be logged

4. Where are Local Policies primarily applied?

- A) In Active Directory environments
- B) On individual computers
- C) On network routers
- D) In cloud environments
- Answer: B) On individual computers

5. What is the difference between Local Policies and Group Policies?

- A) Local Policies apply to networks, while Group Policies apply to single machines
- B) There is no difference
- C) Local Policies are easier to implement
- D) Local Policies apply to single machines, while Group Policies apply to multiple machines in a domain
- Answer: D) Local Policies apply to single machines, while Group Policies apply to multiple machines in a domain

6. Which Local Policy setting allows you to restrict user access to certain applications?

- A) Software Restriction Policies
- B) Audit Policy
- C) User Rights Assignment
- D) Security Options
- Answer: A) Software Restriction Policies

7. What is the function of User Rights Assignment in Local Policies?

- A) Defining what actions users can perform on a computer
- B) Setting password policies
- C) Determining network access levels
- D) Managing software updates
- Answer: A) Defining what actions users can perform on a computer

8. In Local Policies, what is the purpose of Security Options?

- A) To configure firewall settings
- B) To set user account control levels and other security parameters
- C) To assign IP addresses

- D) To manage VPN connections
- Answer: B) To set user account control levels and other security parameters

9. Which Local Policy setting is used to enforce password complexity requirements?

- A) Password Policy
- B) Audit Policy
- C) Account Lockout Policy
- D) User Rights Assignment
- Answer: A) Password Policy

10. Can Local Policies be overridden by Group Policies in a domain environment?

- A) Yes, always
- B) No, never
- C) Only if the Group Policy is set to enforce
- D) Only if the Local Policy is not configured
- Answer: A) Yes, always

Group Policies

1. What is the primary purpose of Group Policy in Windows?

- A) Managing network bandwidth
- B) Configuring and managing computer and user settings in a domain
- C) Providing Internet access
- D) Monitoring system performance
- Answer: B) Configuring and managing computer and user settings in a domain

2. Group Policy settings are processed in which order?

- A) Local, Site, Domain, OU
- B) OU, Domain, Site, Local
- C) Domain, Site, OU, Local
- D) Site, Domain, OU, Local
- Answer: A) Local, Site, Domain, OU

3. What tool is used to create and manage Group Policy Objects (GPOs)?

- A) Group Policy Management Console (GPMC)
- B) Active Directory Users and Computers
- C) Windows PowerShell
- D) System Configuration
- Answer: A) Group Policy Management Console (GPMC)

4. Which type of policy within Group Policy controls the environment of user accounts?

- A) Computer Configuration
- B) User Configuration
- C) Security Settings
- D) Administrative Templates
- Answer: B) User Configuration

5. How are Group Policies applied in an Active Directory environment?

- A) Only manually by an administrator
- B) Automatically when a user logs on or a computer starts up
- C) Through user request
- D) Periodically at fixed intervals
- Answer: B) Automatically when a user logs on or a computer starts up

6. What is a Group Policy Object (GPO) linked to?

- A) A specific user or computer
- B) A network device
- C) An Active Directory container like a Site, Domain, or Organizational Unit
- D) A file server
- Answer: C) An Active Directory container like a Site, Domain, or Organizational Unit

7. What happens when two GPOs with conflicting settings are applied?

- A) The settings cancel each other out
- B) The last applied GPO takes precedence
- C) The first applied GPO takes precedence
- D) An error is generated
- Answer: B) The last applied GPO takes precedence

8. Can Group Policy settings be enforced on computers not joined to a domain?

- A) Yes, using Local Group Policy
- B) No, they only apply to domain-joined computers
- C) Yes, through remote management
- D) Only if the computer is part of a workgroup
- Answer: B) No, they only apply to domain-joined computers

9. In Group Policy, what is the purpose of Administrative Templates?

- A) To manage user accounts
- B) To set network connection settings
- C) To configure registry-based policy settings
- D) To configure security settings
- Answer: C) To configure registry-based policy settings

10. What feature in Group Policy Management allows an administrator to see the potential impact of a GPO before applying it?

- A) Group Policy Modeling
- B) Group Policy Results
- C) Group Policy Update
- D) Group Policy Backup
- Answer: A) Group Policy Modeling

FSMO (Flexible Single Master Operations)

1. Which FSMO role is responsible for managing updates to the schema?

- a) PDC Emulator
- b) Infrastructure Master
- c) Schema Master
- d) Domain Naming Master
- Answer: c) Schema Master

2. Where is the RID Master role located by default?

- a) In each domain
- b) In each site
- c) In the forest root domain
- d) In the global catalog
- Answer: a) In each domain

3. What is the primary function of the Infrastructure Master?

- a) To allocate RIDs
- b) To manage user logins
- c) To update cross-domain references
- d) To replicate schema changes
- Answer: c) To update cross-domain references

4. The PDC Emulator is crucial for what function in a domain?

- a) Time synchronization
- b) Domain name resolution
- c) Schema updates
- d) Global catalog services
- Answer: a) Time synchronization

5. How many Schema Masters can you have in a single forest?

- a) One per forest
- b) One per domain
- c) Multiple per forest
- d) Multiple per domain
- Answer: a) One per forest

6. Which FSMO role is vital for adding or removing domains in a forest?

- a) RID Master
- b) Domain Naming Master
- c) Infrastructure Master
- d) PDC Emulator
- Answer: b) Domain Naming Master

7. If the Infrastructure Master is in the same domain controller as the Global Catalog, what should be done?

- a) Nothing, this is the recommended configuration
- b) Disable the Global Catalog on that domain controller
- c) Move the Infrastructure Master role to another DC
- d) Move the Global Catalog to another DC
- Answer: c) Move the Infrastructure Master role to another DC

8. What happens when a RID pool is depleted?

- a) The domain controller requests a new RID pool
- b) The domain controller restarts
- c) No new objects can be created
- d) The RID Master role is moved to another DC
- Answer: a) The domain controller requests a new RID pool

9. Can FSMO roles be transferred?

- a) Yes, both manually and automatically
- b) No, they are fixed
- c) Yes, but only manually
- d) Yes, but only automatically
- Answer: c) Yes, but only manually

10. What might indicate a problem with the PDC Emulator?

- a) Inability to change domain-wide policies
- b) Slow replication between domain controllers
- c) Inaccurate time across the domain
- d) Failure in allocating new RIDs
- Answer: c) Inaccurate time across the domain

FSRM (File Server Resource Manager)

1. What is the primary function of FSRM?

- a) Monitoring network traffic
- b) Managing disk quotas
- c) Scheduling backups
- d) Allocating RIDs
- Answer: b) Managing disk quotas

2. Which feature of FSRM helps in classifying files for management purposes?

- a) File Screens
- b) Storage Reports
- c) Classification Management
- d) Quota Management
- Answer: c) Classification Management

3. What does a File Screen in FSRM do?

- a) Reports on file usage
- b) Prevents certain file types from being saved
- c) Manages file permissions
- d) Monitors file changes
- Answer: b) Prevents certain file types from being saved

4. What is the purpose of Quota Management in FSRM?

- a) To limit the amount of data a user can store
- b) To monitor network bandwidth usage
- c) To allocate resources for file storage
- d) To set permissions on files
- Answer: a) To limit the amount of data a user can store

5. Which component of FSRM can generate reports on file usage?

- a) File Screens
- b) Storage Reports Management
- c) Quota Management
- d) Classification Management
- Answer: b) Storage Reports Management

6. Can FSRM perform actions based on file classifications?

- a) Yes, including moving files and changing permissions
- b) No, it can only classify files
- c) Yes, but only reporting functions

- d) No, it is only for quota management
- Answer: a) Yes, including moving files and changing permissions

7. **What is the role of the 'File Management Tasks' feature in FSRM?**

- a) To automatically perform tasks based on file properties
- b) To create quotas on file servers
- c) To screen files based on type
- d) To generate reports on file server storage
- Answer: a) To automatically perform tasks based on file properties

8. **Can FSRM send notifications when a quota limit is approached or exceeded?**

- a) Yes, via email or event logs
- b) No, it can only log the event
- c) Yes, but only through event logs
- d) No, FSRM does not support notifications
- Answer: a) Yes, via email or event logs

9. **Which of the following is a benefit of using FSRM?**

- a) Increased network speed
- b) Improved disk space usage
- c) Enhanced file encryption
- d) Better user authentication
- Answer: b) Improved disk space usage

10. **What happens if a user tries to save a file that is blocked by a File Screen?**

- a) The file is saved but marked for review
- b) The user receives an error and the file is not saved
- c) The file is automatically encrypted
- d) The file is redirected to a different folder
- Answer: b) The user receives an error and the file is not saved

Windows Server Backup (WSB)

1. **What is the primary function of Windows Server Backup (WSB)?**

- a) Managing user accounts
- b) Backing up and restoring files, folders, and system state
- c) Monitoring system performance
- d) Installing Windows updates
- Answer: b) Backing up and restoring files, folders, and system state

2. **Which of the following is a limitation of Windows Server Backup?**

- a) Cannot backup to tape drives
- b) Cannot backup individual files or folders
- c) Does not support incremental backups
- d) Requires third-party software to function
- **Answer: a) Cannot backup to tape drives**

3. **In WSB, what is the purpose of the 'Bare Metal Recovery' option?**

- a) To recover system settings only
- b) To restore the entire system, including system state and data volumes
- c) To backup user profiles
- d) To repair corrupted files
- **Answer: b) To restore the entire system, including system state and data volumes**

4. **Which of these is not a feature of Windows Server Backup?**

- a) Scheduling backups
- b) File-level recovery
- c) Real-time monitoring
- d) System state recovery
- **Answer: c) Real-time monitoring**

5. **Can Windows Server Backup be used to backup a system to a network location?**

- a) Yes
- b) No
- c) Only with additional plugins
- d) Only in older versions of Windows Server
- **Answer: a) Yes**

6. **What is the command-line tool for managing Windows Server Backup?**

- a) PowerShell
- b) Wbadmin
- c) Diskpart
- d) BackupExec
- **Answer: b) Wbadmin**

7. **Which of the following cannot be backed up using Windows Server Backup?**

- a) Exchange Server databases
- b) SQL Server databases
- c) FAT32 volumes
- d) NTFS volumes
- **Answer: c) FAT32 volumes**

8. In Windows Server Backup, what does VSS stand for?

- a) Virtual Storage System
- b) Volume Shadow Copy Service
- c) Virtual Server Service
- d) Volume Security System
- **Answer: b) Volume Shadow Copy Service**

9. How can a system be restored using Windows Server Backup?

- a) Through the Control Panel
- b) Using the Windows Recovery Environment
- c) By logging into the system in Safe Mode
- d) Via Remote Desktop
- **Answer: b) Using the Windows Recovery Environment**

10. What file format does Windows Server Backup use for its backups?

- a) .zip
- b) .bak
- c) .vhdx
- d) .wim
- **Answer: c) .vhdx**

Windows Deployment Services (WDS)

1. What is the primary use of Windows Deployment Services (WDS)?

- a) To deploy Windows operating systems over the network
- b) To manage user profiles on Windows systems
- c) To install software updates
- d) To monitor network performance
- **Answer: a) To deploy Windows operating systems over the network**

2. Which of the following is required to use WDS?

- a) Active Directory Domain Services
- b) A DHCP server
- c) A DNS server
- d) All of the above
- **Answer: d) All of the above**

3. In WDS, what is a 'boot image' used for?

- a) To start the client computer and connect to the WDS server
- b) To provide a user interface for selecting different OS images

- c) To store user data during deployment
- d) To install drivers
- **Answer: a) To start the client computer and connect to the WDS server**

4. **What is the role of a 'capture image' in WDS?**

- a) To create a backup of the server
- b) To deploy operating systems
- c) To capture an image of a reference computer for deployment
- d) To provide additional security during deployment
- **Answer: c) To capture an image of a reference computer for deployment**

5. **Which file format does WDS use for storing operating system images?**

- a) .iso
- b) .wim
- c) .vhd
- d) .img
- **Answer: b) .wim**

6. **Can WDS deploy non-Windows operating systems?**

- a) Yes, but with limited functionality
- b) No, it only supports Windows operating systems
- c) Yes, it supports all major operating systems
- d) Only Linux-based operating systems
- **Answer: a) Yes, but with limited functionality**

7. **What protocol does WDS use to transfer data over the network?**

- a) FTP
- b) HTTP
- c) TFTP
- d) SMB
- **Answer: c) TFTP**

8. **Which of the following is a feature of WDS?**

- a) Incremental backups
- b) Multicast transmission
- c) Real-time monitoring
- d) Disk cloning
- **Answer: b) Multicast transmission**

9. **How does WDS integrate with Active Directory?**

- a) It uses Active Directory for user authentication during deployment

- b) It automatically creates computer accounts in Active Directory
- c) It requires Active Directory for storing images
- d) It does not integrate with Active Directory
- **Answer: b) It automatically creates computer accounts in Active Directory**

10. **What is the purpose of the Preboot Execution Environment (PXE) in WDS?**

- a) To provide a command-line interface for the server
- b) To allow client computers to boot from the network and connect to the WDS server
- c) To encrypt data during deployment
- d) To manage user profiles
- **Answer: b) To allow client computers to boot from the network and connect to the WDS server**

IIS Web Server

1. **What does IIS stand for in the context of web servers?**

- a) Integrated Internet Service
- b) Internet Information Services
- c) Internal Interface System
- d) Internet Interface Services
- **Answer: b) Internet Information Services**

2. **Which operating system is required to run IIS Web Server?**

- a) Any version of Windows
- b) Only Windows Server operating systems
- c) Windows 10 and above
- d) Linux-based operating systems
- **Answer: a) Any version of Windows**

3. **What is the primary use of IIS Web Server?**

- a) To host websites and web applications
- b) To manage Windows updates
- c) To serve as a file server
- d) To monitor network traffic
- **Answer: a) To host websites and web applications**

4. **Which of the following is a feature of IIS?**

- a) Support for multiple programming languages
- b) Automatic backup of web content
- c) Built-in antivirus protection
- d) Real-time user monitoring

- **Answer: a) Support for multiple programming languages**

5. **How can the performance of IIS Web Server be monitored?**

- a) Using Windows Task Manager
- b) Through IIS Manager
- c) Via command-line tools
- d) Both b and c
- **Answer: d) Both b and c**

6. **What is the purpose of Application Pools in IIS?**

- a) To allocate resources to different websites or applications
- b) To pool user connections
- c) To manage network bandwidth
- d) To provide additional security
- **Answer: a) To allocate resources to different websites or applications**

7. **Which of these file types does IIS use to store its configuration settings?**

- a) .xml
- b) .ini
- c) .config
- d) .json
- **Answer: c) .config**

8. **Can IIS host both static and dynamic web content?**

- a) Yes
- b) No
- c) Only static content
- d) Only dynamic content
- **Answer: a) Yes**

9. **Which protocol is primarily used by IIS to transfer web content?**

- a) FTP
- b) HTTP/HTTPS
- c) SMB
- d) TCP/IP
- **Answer: b) HTTP/HTTPS**

Exchange Server

1. **What is the primary function of Microsoft Exchange Server?**

- a) Database management
- b) Email communication
- c) Web hosting
- d) Virtualization
- **Answer:** b) Email communication

2. Which protocol is used by Exchange Server for mail synchronization on mobile devices?

- a) IMAP
- b) POP3
- c) SMTP
- d) ActiveSync
- **Answer:** d) ActiveSync

3. What is the purpose of a Database Availability Group (DAG) in Exchange Server?

- a) Load balancing
- b) Data replication
- c) User authentication
- d) Spam filtering
- **Answer:** b) Data replication

4. Which feature in Exchange Server is used for preventing data loss?

- a) DAG
- b) Transport rules
- c) Data Loss Prevention (DLP)
- d) Mailbox auditing
- **Answer:** c) Data Loss Prevention (DLP)

5. What is the PowerShell cmdlet to create a new mailbox in Exchange Server?

- a) New-Mailbox
- b) Create-Mailbox
- c) Add-Mailbox
- d) Set-Mailbox
- **Answer:** a) New-Mailbox

6. Which Exchange Server role handles client connections for mail access?

- a) Mailbox Server Role
- b) Client Access Server Role
- c) Edge Transport Server Role
- d) Hub Transport Server Role
- **Answer:** b) Client Access Server Role

7. What is the use of the Edge Transport Server role in Exchange Server?

- a) Mailbox storage
- b) Internal mail routing
- c) Internet mail flow and security
- d) Client connectivity
- **Answer:** c) Internet mail flow and security

8. Which tool is used for Exchange Server deployment and troubleshooting?

- a) Exchange Admin Center
- b) Exchange Management Console
- c) Exchange Control Panel
- d) Exchange Management Shell
- **Answer:** d) Exchange Management Shell

9. What is the role of the Global Address List (GAL) in Exchange Server?

- a) Store emails
- b) Maintain user mailbox settings
- c) Provide a centralized address list
- d) Filter spam emails
- **Answer:** c) Provide a centralized address list

10. How does Exchange Server support high availability and disaster recovery?

- a) Through single-instance storage
- b) Using Windows Server Backup
- c) By implementing DAGs
- d) Through SMTP relay
- **Answer:** c) By implementing DAGs

Network Load Balancing (NLB)

1. What is the main purpose of Network Load Balancing (NLB)?

- a) Increase security
- b) Manage user accounts
- c) Distribute traffic across multiple servers
- d) Data backup
- **Answer:** c) Distribute traffic across multiple servers

2. Which layer of the OSI model does NLB operate at?

- a) Transport layer
- b) Network layer

- c) Session layer
- d) Application layer
- **Answer:** b) Network layer

3. **What is a primary benefit of using NLB in a server cluster?**

- a) Data encryption
- b) High availability
- c) Centralized management
- d) Cost reduction
- **Answer:** b) High availability

4. **Which Windows feature needs to be installed to use NLB?**

- a) Hyper-V
- b) Windows Defender
- c) Network Load Balancing feature
- d) Remote Desktop Services
- **Answer:** c) Network Load Balancing feature

5. **How does NLB ensure availability if one server fails?**

- a) By automatically restarting the server
- b) Through server replication
- c) By redistributing traffic to remaining servers
- d) By generating error reports
- **Answer:** c) By redistributing traffic to remaining servers

6. **What type of IP address is used by NLB clusters for client connections?**

- a) Static IP address
- b) Virtual IP address
- c) Dynamic IP address
- d) Private IP address
- **Answer:** b) Virtual IP address

7. **What is the maximum number of nodes supported in a Windows NLB cluster?**

- a) 2
- b) 8
- c) 16
- d) 32
- **Answer:** d) 32

8. **What is a port rule in NLB?**

- a) Security setting for ports

- b) Defines how traffic is distributed to servers
- c) Assigns specific ports to nodes
- d) Manages port security
- **Answer:** b) Defines how traffic is distributed to servers

9. **Which type of traffic can NLB not be used for?**

- a) HTTP
- b) FTP
- c) UDP
- d) Multicast
- **Answer:** d) Multicast

10. **In NLB, what is "convergence"?**

- a) Process of balancing the load
- b) The initial setup of the cluster
- c) The process where cluster configuration updates
- d) Monitoring the performance
- **Answer:** c) The process where cluster configuration updates

Security Issues in Active Directory

1. **What is a common security concern in Active Directory?**

- a) Insufficient disk space
- b) Unauthorized access to data
- c) Network speed
- d) Printer configuration
- **Answer:** b) Unauthorized access to data

2. **What does LDAP signing in Active Directory do?**

- a) Increases processing speed
- b) Reduces data storage needs
- c) Prevents tampering with data in transit
- d) Compresses data
- **Answer:** c) Prevents tampering with data in transit

3. **Which feature in Active Directory helps in delegating specific administrative tasks?**

- a) Organizational Units
- b) Group Policies
- c) Security groups
- d) Active Directory Federation Services

- **Answer:** a) Organizational Units

4. **What is the purpose of a Read-Only Domain Controller (RODC)?**

- a) Improves write performance
- b) Provides a backup of the domain controller
- c) Hosts a read-only copy of the domain database
- d) Manages printer resources
- **Answer:** c) Hosts a read-only copy of the domain database

5. **How does Active Directory use Group Policy for security?**

- a) By managing user accounts
- b) By configuring security settings for users and computers
- c) By organizing domain objects
- d) By providing email services
- **Answer:** b) By configuring security settings for users and computers

6. **What is the Kerberos protocol used for in Active Directory?**

- a) Data encryption
- b) User authentication
- c) Data backup
- d) Network load balancing
- **Answer:** b) User authentication

7. **Which is a best practice for securing Active Directory?**

- a) Regularly updating hardware
- b) Using complex passwords
- c) Reducing the number of printers
- d) Increasing storage capacity
- **Answer:** b) Using complex passwords

8. **What is a potential security risk when using service accounts in Active Directory?**

- a) High resource usage
- b) Insufficient data storage
- c) Privilege escalation
- d) Network congestion
- **Answer:** c) Privilege escalation

9. **What is the purpose of a Security Descriptor in Active Directory?**

- a) Defines storage limits
- b) Manages network traffic
- c) Sets security permissions on objects

- d) Organizes domain controllers
- **Answer:** c) Sets security permissions on objects

10. **How can administrators monitor for potential security breaches in Active Directory?**

- a) By using Performance Monitor
- b) By reviewing security logs
- c) By conducting user surveys
- d) By increasing server memory
- **Answer:** b) By reviewing security logs

Windows PowerShell MCQs

1. **What is the primary purpose of Windows PowerShell?**

- A) Web browsing
- B) Text processing
- C) Automation and scripting
- D) Image editing
- **Answer: C) Automation and scripting**

2. **Which cmdlet is used to get help about a specific command in PowerShell?**

- A) Get-Help
- B) Get-Command
- C) Get-Info
- D) Get-Support
- **Answer: A) Get-Help**

3. **What does the cmdlet Get-Process do in PowerShell?**

- A) Installs a new process
- B) Retrieves information about running processes
- C) Ends a process
- D) Creates a new process
- **Answer: B) Retrieves information about running processes**

4. **How do you create a new directory using PowerShell?**

- A) mkdir
- B) mkdir
- C) New-Directory
- D) New-Item
- **Answer: D) New-Item**

5. Which cmdlet is used to list all services on your machine?

- A) Get-Services
- B) List-Services
- C) Show-Services
- D) Get-Service
- **Answer: D) Get-Service**

6. What is a 'cmdlet' in PowerShell?

- A) A command-line interface
- B) A lightweight script that performs a specific function
- C) A module containing multiple scripts
- D) A configuration file
- **Answer: B) A lightweight script that performs a specific function**

7. How do you execute a PowerShell script file?

- A) .\script.ps1
- B) run script.ps1
- C) execute script.ps1
- D) script.ps1
- **Answer: A) .\script.ps1**

8. What is the default scripting language used in PowerShell?

- A) JavaScript
- B) Python
- C) PowerShell Script (PS1)
- D) Batch
- **Answer: C) PowerShell Script (PS1)**

9. Which cmdlet is used to copy a file in PowerShell?

- A) Copy-File
- B) Copy-Item
- C) Move-Item
- D) Transfer-File
- **Answer: B) Copy-Item**

10. How do you display the contents of a text file in PowerShell?

- A) show-content
- B) display-file
- C) cat
- D) Get-Content

- **Answer: D) Get-Content**

11. **What does the pipeline symbol | do in PowerShell?**

- A) Divides commands
- B) Adds multiple commands
- C) Passes the output of one cmdlet as input to another cmdlet
- D) Concatenates commands
- **Answer: C) Passes the output of one cmdlet as input to another cmdlet**

12. **Which cmdlet is used to view the history of commands executed in the current session?**

- A) Get-History
- B) View-History
- C) Show-History
- D) History-Get
- **Answer: A) Get-History**

13. **How do you stop a running script or command in PowerShell?**

- A) Stop-Command
- B) Ctrl + C
- C) End-Script
- D) Break
- **Answer: B) Ctrl + C**

14. **Which cmdlet is used to change the current working directory in PowerShell?**

- A) Change-Directory
- B) GoTo-Directory
- C) Set-Location
- D) Move-Location
- **Answer: C) Set-Location**